



[비용기반 임계값 설정](1차 전처리)



Task : 비용기반 임계값 설정

- 위험 점수(risk) = $1 - P(\text{정상 } m)$
- if risk $\geq$ 임계값 = "사기 의심"으로 판정 $\Rightarrow a \sim i$ 중 확률 최고인 유형으로 분류
 - FN = 실제로는 사기인데, 모델이 정상이라고 판정 $\Rightarrow$ 실제로 사기인데 탐지를 못했으므로 사기 피해액이 발생
 - FP = 실제로는 정상인데, 모델이 사기라고 판정 $\Rightarrow$ 실제로 정상인데 사기로 탐지를 하면 정상고객의 불만 대응 및 탈퇴이슈가 발생함

- FN/FP(비용 기반 임계값)으로 임계값 설정

2023년 기준 보이스 피싱 피해액 약 1965억원 / 1인당 약 1700만원 = FN단가

FP비용 : 5000원 = 피해 방지를 최우선 / 1만원 = 오탐 고객 응대시 기본값 예상 / 5만원 = 오탐 민원/이탈 비용 크게 생각

보도자료(상세) | 보도자료 | 보도·알림 |

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 <https://www.fss.or.kr/fss/bbs/B0000188/view.do?nttlId=134451&menuNo=200218>



금융감독원
FINANCIAL SUPERVISORY SERVICE

시나리오	FN 단가 (사기 1건 놓쳤을 때 피해액)	FP 단가 (정상 1건 잘못 막았을 때 비용)
A.	1700만원	5000원
B.	1700만원	1만원
C.	1700만원	5만원

LightGBM(한방) — 임계값별 기대비용 (단위: 원, FN단가 1,700만원)

임계값	FN	FP	비용 A(FP 5천원)	비용 B(FP 1만원)	비용 C(FP 5만원)
0.9	68	1	1,156,005,000	1,156,010,000	1,156,050,000

임계값	FN	FP	비용 A(FP 5천원)	비용 B(FP 1만원)	비용 C(FP 5만원)
0.8	60	2	1,020,010,000	1,020,020,000	1,020,100,000
0.7	53	3	901,015,000	901,030,000	901,150,000
0.6	51	3	867,015,000	867,030,000	867,150,000
0.5	48	4	816,020,000	816,040,000	816,200,000
0.4	44	4	748,020,000	748,040,000	748,200,000
0.3	42	4	714,020,000	714,040,000	714,200,000
0.2	38	5	646,025,000	646,050,000	646,250,000
0.1	35	7	595,035,000	595,070,000	595,350,000
0.09	34	7	578,035,000	578,070,000	578,350,000
0.08	32	8	544,040,000	544,080,000	544,400,000
0.07	32	8	544,040,000	544,080,000	544,400,000
0.06	31	8	527,040,000	527,080,000	527,400,000
0.05	31	9	527,045,000	527,090,000	527,450,000
0.04	30	10	510,050,000	510,100,000	510,500,000
0.03	27	12	459,060,000	459,120,000	459,600,000
0.02	27	13	459,065,000	459,130,000	459,650,000
0.01	24	20	408,100,000	408,200,000	409,000,000
0.005	17	23	289,115,000	289,230,000	290,150,000
0.001	12	50	204,250,000	204,500,000	206,500,000
0.000	0	23,602	27,015,000 (A 최저·붕괴)	54,030,000 (B 최저·붕괴)	270,150,000

LightGbm(임계값O)

임계값 (미탐/오탐 트레이드오프)

임계값	Macro-F1	FN	FP
0.900	0.6187	68	1
0.800	0.6448	60	2
0.700	0.6632	53	3
0.600	0.6653	51	3
0.500	0.6684	48	4
0.400	0.6801	44	4
0.300	0.6818	42	4
0.200	0.6926	38	5
0.150	0.6976	36	5
0.100	0.6982	35	7

임계값	Macro-F1	FN	FP
0.090	0.6973	34	7
0.080	0.7003	32	8
0.070	0.7003	32	8
0.060	0.7028	31	8
0.050	0.7009	31	9
0.040	0.6987	30	10
0.030	0.7057	27	12
0.020	0.7039	27	13
0.010	0.6995	24	20
0.007	0.7087	19	21
0.005	0.7103	17	23
0.003	0.7013	16	32
0.001	0.6857	12	50
0.000	0.0219	0	23,602

임계값 적용 & 지표

LightGBM_임계값 0.005_유형별

[임계값 0.005] Macro-F1=0.7123 | Accuracy=0.9956 | 사기검출 Recall=0.9186

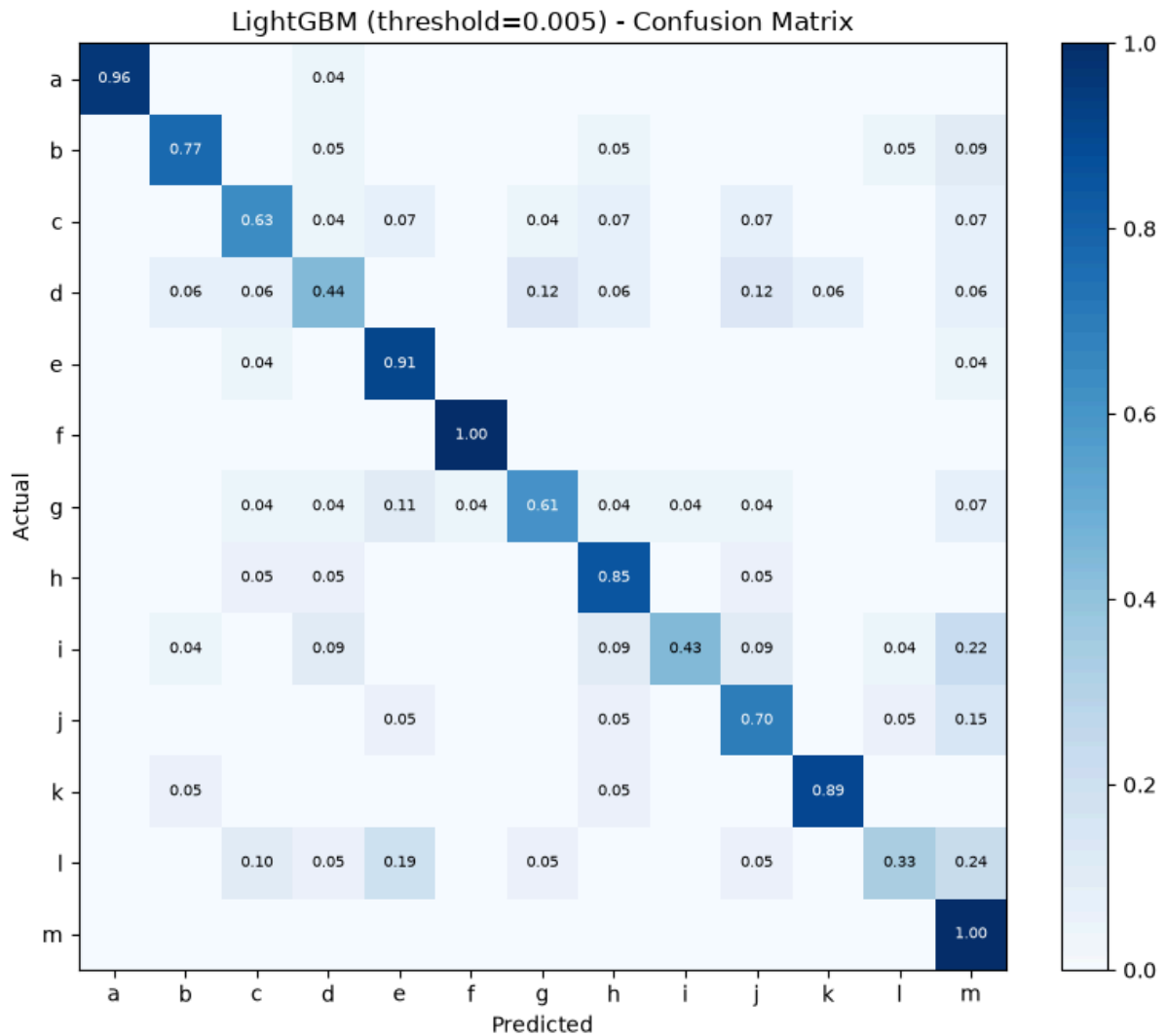
[클래스별 성능] (a~l=사기유형, m=정상)

precision recall f1-score support

a	1.00	0.96	0.98	23
b	0.85	0.77	0.81	22
c	0.71	0.63	0.67	27
d	0.47	0.44	0.45	16
e	0.58	0.91	0.71	23
f	0.73	1.00	0.84	16
g	0.81	0.61	0.69	28
h	0.63	0.85	0.72	20
i	0.56	0.43	0.49	23
j	0.61	0.70	0.65	20
k	0.89	0.89	0.89	19
l	0.37	0.33	0.35	21
m	1.00	1.00	1.00	23602
accuracy			1.00	23860

macro avg 0.71 0.73 0.71 23860

weighted avg 1.00 1.00 1.00 23860



[유형별 Feature Importance] 유형마다 이진 모델 학습 중 ...

[유형별 상위 중요 피쳐 top3]

a: Time_difference_seconds, Customer_Birthyear, Account_initial_balance

b: Transaction_Amount, Transaction_Amount_abs, Customer_credit_rating

c: terminal_malicious_count, amount_to_month_max_ratio, Transaction_Amount

d: amount_to_month_max_ratio, Transaction_Amount, days_since_last_atm

e: Transaction_Amount, days_since_last_atm, balance_depletion_ratio

f: balance_depletion_ratio, Channel, Transaction_Amount

g: Transaction_Amount, balance_depletion_ratio, amount_to_daily_limit_ratio

h: amount_to_month_max_ratio, Account_one_month_max_amount, Transaction_Amount

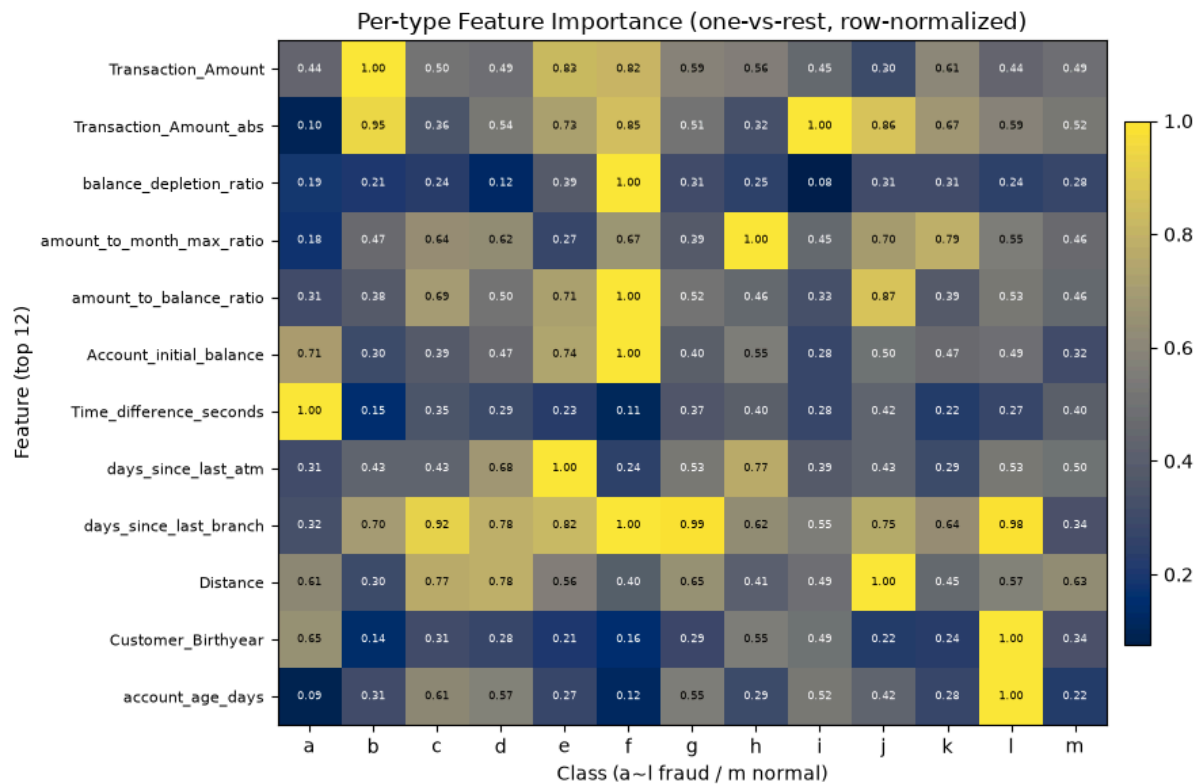
i: Account_dawn_one_month_max_amount, Account_dawn_one_month_std_dev, Transaction_Amount_abs

j: Transaction_Amount_abs, Account_account_type, amount_to_balance_ratio

k: Channel, amount_to_month_max_ratio, Transaction_Amount

l: Customer_Birthyear, account_age_days, Transaction_Amount_abs

m: Transaction_Amount, Transaction_Amount_abs, Account_account_type



[유형별 SHAP] TreeExplainer 계산 (표본 2000행) ...

[유형별 상위 기여 피쳐 top3]

a: Time_difference_seconds, Distance, Transaction_Amount

b: Customer_credit_rating, device_network_risk_count, Channel

c: terminal_malicious_count, Channel, Transaction_num_connection_failure

d: Channel, Access_Medium, Transaction_Amount

e: days_since_last_atm, Channel, Flag_deposit_more_than_tenMillion

f: days_since_last_branch, Flag_deposit_more_than_tenMillion, days_since_resumed

g: Flag_deposit_more_than_tenMillion, Account_release_suspension, Transaction_Amount_abs

h: Account_one_month_std_dev, Account_one_month_max_amount, amount_to_month_max_ratio

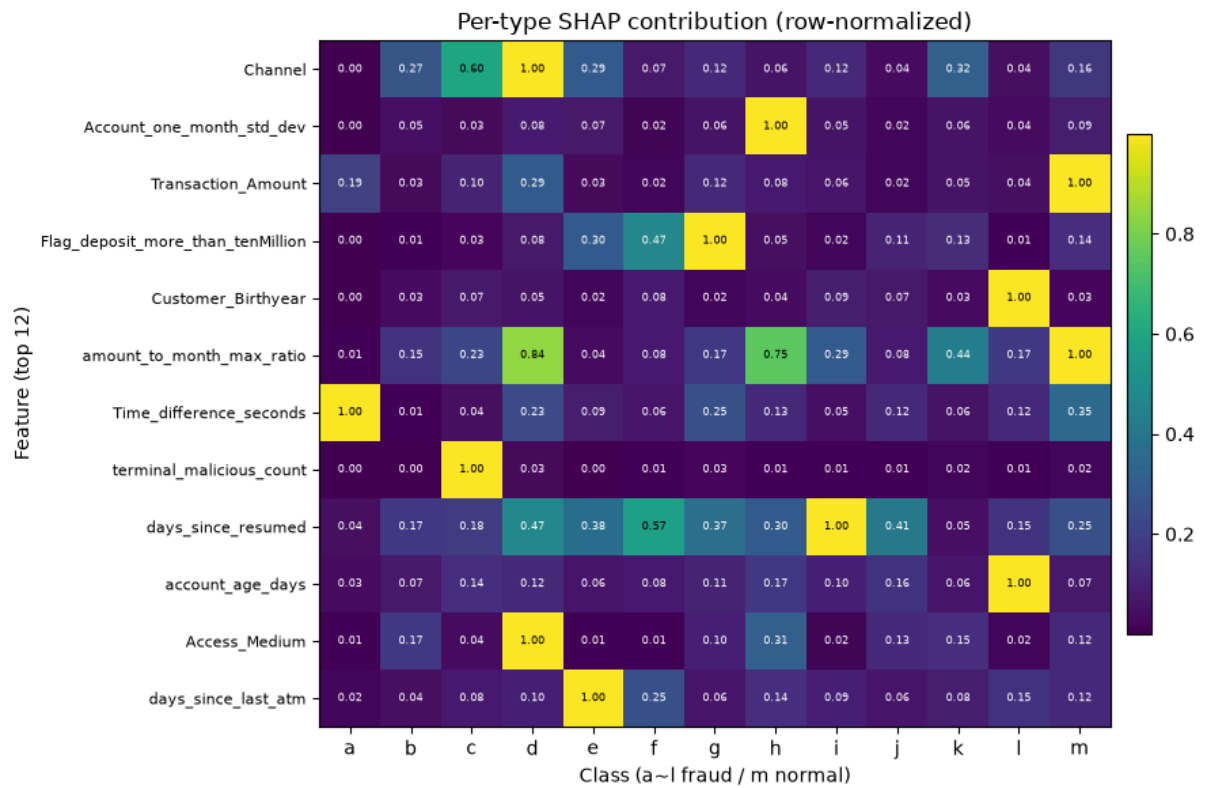
i: Account_dawn_one_month_max_amount, Account_dawn_one_month_std_dev, days_since_resumed

j: suspended_recipient_and_withdrawal, Transaction_history_with_the_account, Account_account_type

k: Number_of_transaction_with_the_account, Channel, amount_to_month_max_ratio

l: Customer_Birthyear, account_age_days, Customer_loan_type

m: Transaction_Amount, Transaction_is_withdrawal, amount_to_month_max_ratio



실행 및 진행 사항 정리



결과